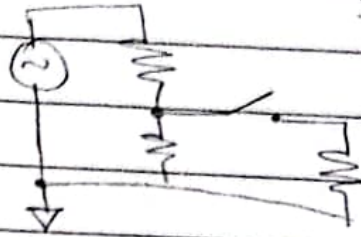




Mid Exam Preparation



KCL:

Incoming currents are equal to outgoing currents

$$I_1 + I_2 = I_3 + I_4$$

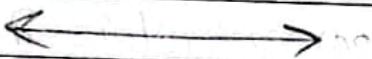


VDR:

If two resistors are connected in series then

$$V_1 = V_{in} \frac{R_1}{R_1 + R_2}$$

$$V_2 = V_{in} \frac{R_2}{R_1 + R_2}$$



Node Voltage:

Based on KCL, choose a reference Node, and then write current in terms of Ohm's Law.

$$\frac{V_s - V_1}{R_1} + \frac{0 - V_1}{R_2} = 0$$



CDR:

If two resistors are connected in parallel then

$$I_1 = I_{total} \times \frac{R_2}{R_1 + R_2}$$

$$I_2 = I_{total} \times \frac{R_1}{R_1 + R_2}$$



Mesh Analysis:

Based on KVL, just written in form of Ohm's Law.

Mesh 1:

$$-V_s + I_1 R_1 + (I_1 - I_2) R_3 = 0$$

Mesh 2:

$$(I_2 - I_1) R_3 + I_2 R_2 = 0$$

KVL:

Sum of all voltages equal to zero.

$$V - V_1 - V_2 = 0$$

$$V = V_1 + V_2$$

